

Malaysia's Semiconductor Ascent: From OSAT Colony to Intelligent Manufacturing Hub

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Executive Summary

Malaysia stands at the most consequential inflection point in its fifty-four-year semiconductor history. The country that began as a low-cost assembly destination for Intel in 1972 now controls 13% of the global chip assembly and testing market, ranks as the world's seventh-largest semiconductor exporter, and is positioning itself — with RM25 billion in government fiscal support and RM500 billion in targeted private investment — to ascend the value chain from pure back-end outsourcer to integrated packaging innovator and, eventually, a credible front-end manufacturing jurisdiction. This report examines the architecture of that ambition: the National Semiconductor Strategy, Intel's "Project Pelican," the compound semiconductor opportunity, the Johor-Singapore SEZ as a structural catalyst, the talent crisis that threatens to derail everything, and the geopolitical crossfire between Washington and Beijing that gives every policy decision existential weight.

I. The Paradox of Malaysia's Chip Industry

Malaysia is the world's seventh-largest semiconductor exporter — a statement that most Western semiconductor strategy discussions treat as a footnote rather than a fact that demands explanation. In Washington and Brussels, the semiconductor conversation is dominated by Taiwan (TSMC), South Korea (Samsung, SK Hynix), Japan (legacy fabs, advanced materials), and the United States itself (Intel, Qualcomm, Texas Instruments, Applied Materials). Malaysia rarely appears in this hierarchy despite holding a larger share of the global chip assembly and testing market than Germany, France, or the United Kingdom.

The invisibility is structural, not accidental. Malaysia's semiconductor identity is built on the back-end: the unglamorous but operationally indispensable work of assembling, packaging, and

testing chips after the wafers have been cut at high-prestige fabs in Taiwan and South Korea. "Back-end" is a descriptor that inadvertently communicates hierarchy — a country at the back, trailing, subordinate. It obscures the reality that without reliable, high-volume, high-precision packaging and test services, every cutting-edge chip designed in Silicon Valley and fabricated in Hsinchu remains a wafer, not a product.

Now, in 2026, that self-effacing identity is under active revision. Malaysia's Prime Minister Anwar Ibrahim launched the National Semiconductor Strategy (NSS) in May 2024, backed by RM25 billion in government funding and a RM500 billion private investment target over a decade. Intel is completing its largest 3D chip packaging facility on Earth — Project Pelican — in Penang, valued at approximately \$7 billion. CHIPX, an Irish-Taiwanese compound semiconductor startup, has announced the ASEAN region's first 8-inch GaN-on-SiC wafer fabrication facility on Malaysian soil. Arm Holdings has executed a RM1.1 billion national partnership providing 32 semiconductor design licenses to Malaysian companies. The Johor-Singapore Special Economic Zone, launched in January 2025, is creating a new geography for advanced manufacturing that exploits Singapore's intellectual property and legal infrastructure alongside Malaysia's land, labour, and infrastructure advantages.

The question is not whether Malaysia's semiconductor ambitions are real — they demonstrably are, backed by capital commitments and construction sites. The question is whether they are achievable at the timeline and scale the government has specified, against a background of structural talent shortages, geopolitical pressure from both Washington and Beijing, and the brutal capital intensity of true front-end semiconductor manufacturing.

This report provides the first systematic intelligence assessment of Malaysia's semiconductor trajectory in 2026, drawing on live industry data, government strategy documents, and primary source reporting.

Sources:

- Malaysia's semiconductor journey spanning half a century — EDN Asia
- Malaysia Eyes Nearly \$200 Billion in Electronics Exports in 2026 — Bloomberg

II. Malaysia's Semiconductor Heritage: From Free Trade Zone to Global Back-End Capital

The origin story of Malaysia's semiconductor industry begins in Penang in 1972, and it begins with Intel.

Robert Noyce and Gordon Moore's company, then barely four years old, was already confronting the economics of semiconductor manufacturing: the chips were getting smaller and cheaper, but the assembly work — bonding die to lead frames, encapsulating in resin, testing for yield — remained stubbornly labour-intensive. The solution was geographic arbitrage. In 1972, Intel opened a 5-acre assembly plant in Bayan Lepas, Penang. It employed nearly a thousand

workers. By 1975, that single facility in Malaysia accounted for more than half of Intel's global assembly capacity. The economics were unambiguous.

AMD followed Intel to Penang. Then Hitachi, Hewlett-Packard, National Semiconductor, Motorola. By the early 1980s, fourteen semiconductor firms were operating in Malaysia, nearly all of them back-end assembly operations clustered in Penang's free trade zones. The Malaysian government, under Prime Minister Mahathir's industrialisation agenda, actively courted these multinationals with tax incentives, streamlined customs for electronics components, and a policy of building dedicated industrial parks — most consequentially, Kulim Hi-Tech Park in Kedah, established in 1996, which would become the country's premier semiconductor industrial estate.

Over five decades, what began as a single Intel assembly plant in Penang evolved into a dense ecosystem of the world's largest semiconductor companies operating back-end manufacturing at scale. The roster of multinationals active in Malaysia today reads as a who's-who of global semiconductors: Intel, AMD, Infineon, STMicroelectronics, Renesas, Texas Instruments, NXP Semiconductors (the successor to Motorola and Freescale's semiconductor business), ON Semiconductor, ASE Group (the world's largest OSAT company), and Amkor Technology. Infineon alone has expanded to three wafer fab modules at Kulim, making it one of the company's most significant global manufacturing sites.

The Malaysian-owned semiconductor players — smaller in scale, but strategically important as national industrial assets — include four listed companies that together represent the country's indigenous manufacturing base:

Inari Amertron Berhad is Malaysia's largest listed semiconductor company, specialising in radio-frequency (RF) semiconductor assembly and testing. Its primary customer base is Apple's supply chain via Broadcom, making it unusually exposed to iPhone product cycles.

Unisem (M) Berhad provides assembly and test services for automotive, industrial, and communications semiconductors, with operations in Malaysia, Indonesia, and China. It was acquired by a Chinese consortium in 2018, creating strategic ambiguity about its role in a bifurcating technology landscape.

Globetronics Technology Berhad, headquartered in Penang, focuses on LED, sensor, and specialty device packaging. Revenue has declined from approximately RM180 million in 2022 to around RM131 million in 2023 as smartphone sensor demand contracted, illustrating the cyclical vulnerability of pure-OSAT business models.

Silterra Malaysia Sdn Bhd is the critical outlier: Malaysia's only significant domestic wafer foundry, operating an 8-inch facility at Kulim Hi-Tech Park with process nodes between 110nm and 180nm. Silterra is a strategic investment of Dagang NeXchange Berhad (DNeX) and a Chinese private equity fund (CGP Fund), operating at mature nodes primarily serving automotive, power management, and IoT device manufacturers.

Together, these players have constructed a back-end manufacturing ecosystem that today accounts for approximately 13% of the global semiconductor assembly, packaging, and test market — a figure that has remained remarkably stable, and that the NSS explicitly aims to defend and upgrade rather than cede. Malaysia is the world's seventh-largest semiconductor exporter overall. Electronics — of which semiconductors form the dominant component, approximately 65% by value — represent the country's single largest export category, potentially exceeding RM800 billion (approximately USD197 billion) in 2026, according to the Malaysian Electrical and Electronic Industry Association.

The durability of Malaysia's OSAT franchise across five decades reflects genuine competitive advantages: a trained workforce familiar with cleanroom discipline, an established supply chain for packaging materials and chemicals, proximity to Singapore's financial and logistics infrastructure, English-language business environment, and a government that has consistently prioritised the semiconductor sector as a national economic pillar. It also reflects a strategic trap: the same low-cost, high-volume assembly model that attracted Intel in 1972 is under pressure in 2026 from rising Malaysian wages, aggressive competition from Vietnam and Thailand, and the structural shift in semiconductor value creation toward the front-end and design layers that Malaysia has historically ceded to others.

Sources:

- International Expansion for Assembly — Intel History Vault
 - Malaysia's semiconductor journey spanning half a century — EDN
 - Silterra Malaysia — Wikipedia
 - DNeX and Foxconn's subsidiary to build and operate a new 12-inch wafer fabrication plant — Dagang Net
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III. The National Semiconductor Strategy: RM25 Billion and Five Hard Targets

On 28 May 2024, Prime Minister Anwar Ibrahim stood before an audience of semiconductor industry executives and formally launched Malaysia's National Semiconductor Strategy — the most ambitious industrial policy document in the country's history.

The NSS is structured around five headline targets, each with specific numeric commitments that provide a measurable basis for evaluating execution:

Target 1: RM500 billion in cumulative investment. The government aims to attract a minimum of RM500 billion in combined foreign and domestic direct investment in the semiconductor sector across a decade. Foreign direct investment (FDI) would be directed primarily toward wafer fabs and manufacturing equipment; domestic direct investment toward IC design, advanced packaging, and equipment manufacturing. By early 2025, the NSS had already attracted RM54.2 billion in its first year — a strong start, though still at roughly 10.8% of

the decade target in year one. Malaysia's total approved investments reached a record RM426.7 billion in 2025 across all sectors, up 11% year-on-year, with semiconductors as a primary driver.

Target 2: Large-scale local champions. The strategy calls for establishing at least 10 Malaysian-owned companies in IC design and advanced packaging with revenues between RM1 billion and RM4.7 billion. A second tier targets at least 100 semiconductor-related companies with revenues approaching RM1 billion. This is a structural attempt to transform Malaysia's semiconductor industry from a largely foreign-dominated ecosystem into one with genuine indigenous anchors — a deliberate replication of Taiwan's model, where TSMC and its ecosystem suppliers are domestically owned even if globally integrated.

Target 3: 60,000 high-skilled engineers. The human capital target is, in the government's own framing, the foundation for everything else. Malaysia commits to training and upskilling 60,000 high-skilled semiconductor engineers and specialists to support the industry. Given the current gap — industry estimates suggest Penang alone is short approximately 50,000 engineers — this is less an aspiration than a minimum survival condition for the NSS to function at all.

Target 4: RM25 billion in government fiscal support. The government allocated RM25 billion (approximately USD5.3 billion) to operationalise the NSS. This fiscal commitment covers incentives, infrastructure development, training programmes, and co-investment in strategic projects. For context, the US CHIPS and Science Act allocated USD52.7 billion; the EU Chips Act EUR43 billion; Japan's semiconductor stimulus JPY4 trillion. Malaysia's RM25 billion is modest by absolute scale but significant relative to the size of the economy and represents the largest single industrial policy commitment in Malaysian history.

Target 5: Value chain ascent. The NSS explicitly targets Malaysia's progression across all four semiconductor value chain stages: front-end manufacturing (wafer fabrication), mid-end manufacturing (die preparation and advanced packaging), back-end manufacturing (final assembly and test, the current stronghold), and IC design (the high-value creative layer). The three-phase structure spans a decade: Phase 1 modernises OSAT, attracts equipment manufacturers, and grows existing fabs; Phase 2 builds indigenous front-end capability and scales IC design; Phase 3 positions Malaysia as a complete-stack semiconductor jurisdiction.

Two infrastructure anchors underpin the physical realisation of the NSS. **Kulim Hi-Tech Park Phase 3** — the northern expansion of the existing Kulim industrial estate in Kedah — is designated as the primary site for new fab investments and equipment manufacturers. It is here that Infineon has been expanding, that Intel's "Project Pelican" packaging complex rises, and where the DNeX-Foxconn 12-inch wafer fab joint venture is eventually planned. **Penang Science Park II** in Batu Kawan, on the mainland across from Penang Island, is designated for R&D, equipment testing, and semiconductor support industries. The Malaysia Semiconductor IC Design Park 2, launched in Cyberjaya in November 2025, provides the third anchor specifically for IC design activities, equipped with chip design emulation tools, prototype development facilities, and an Advanced Chip Testing Centre.

The New Incentive Framework (NIF), which took effect on 1 March 2026 and supersedes the previous Pioneer Status and Investment Tax Allowance regime, represents the updated fiscal architecture for attracting semiconductor investment. It is designed to reward high-value activities — specifically chip design, advanced packaging, and wafer fabrication — more generously than commodity assembly.

The NSS has attracted significant early momentum. By Q1 2026, Malaysia secured RM92.8 billion in total approved investments for the quarter, with Japan emerging as the largest foreign investor. Within semiconductor-specific FDI, the AIXTRON SE facility announced in May 2026 — a RM47 million greenfield semiconductor equipment manufacturing plant at Bandar Cassia Technology Park, Penang — represents the new type of investment the NSS is explicitly targeting: not just chip assemblers, but the equipment makers who supply the fabs.

Sources:

- Govt allocates RM25bil to operationalise National Semiconductor Strategy — MIDA
 - PM launches National Semiconductor Strategy, outlines 5 targets — FMT
 - Malaysia Breaks Investment Record with RM426.7 Billion in 2025 — MIDA
 - malaysia attracts rm92.8 billion in q1 2026 — Bernama
 - NSS First Year — TNGlobal
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IV. Intel's Kulim Gambit: The World's Largest 3D Packaging Complex

The single most consequential semiconductor investment commitment in Malaysia's history is Intel's "Project Pelican" — an advanced packaging complex that is now completing construction in Penang and is valued by Intel at approximately \$7 billion.

The scale deserves emphasis. Project Pelican is not a chip fabrication plant in the conventional front-end sense — it does not grow silicon crystals, deposit transistors, or perform photolithography at sub-10nm nodes. It is, instead, the world's largest facility dedicated to a specific and rapidly ascending technology: 3D advanced packaging, specifically Intel's proprietary EMIB (Embedded Multi-Die Interconnect Bridge) and Foveros technologies. By 2028, Intel plans the facility to handle packages measuring 120 x 180 millimetres — enormous by conventional chip standards — capable of integrating as many as twenty-four HBM (High Bandwidth Memory) stacks alongside compute dies. The facility is designed to handle die sort, die prep, and full production flows across both EMIB and Foveros processes.

As of mid-2026, Project Pelican is reported at 99% construction completion. Intel will inject an additional \$208 million to finalise the site and bring first-phase operations online. HBM shipments from SK Hynix and Micron have already begun flowing to the Malaysia facility, corroborating that production ramp is underway rather than merely imminent. A separate but related development emerged in August 2026: TSMC — Intel's primary competitor in foundry

services — was reported to be exploring use of Intel's Malaysia Pelican facility for CoWoS (Chip on Wafer on Substrate) back-end capacity support, a remarkable indicator of how advanced packaging is becoming an industry-wide infrastructure layer rather than a company-specific competitive asset.

The strategic rationale behind Intel's decision to site its largest packaging complex in Malaysia rather than Arizona, Ireland, or Germany reflects several converging factors:

Labour cost and availability. Skilled packaging engineers in Malaysia command significantly lower salaries than equivalent personnel in the United States or Germany, and in 2022 Malaysia's engineering workforce depth was adequate at the scale Intel required. This dynamic is under pressure — the talent shortage now is acute — but the initial investment decision was taken when the labour cost advantage was clearer.

Supply chain proximity. Malaysia sits at the centre of the global semiconductor assembly supply chain, with chemical suppliers, lead frame manufacturers, substrate producers, and test equipment service providers within a few hundred kilometres. Moving advanced packaging to Malaysia shortens the supply chain between die preparation and final device delivery to customers in Asia.

Intel's IFS (Intel Foundry Services) angle. Intel is competing with TSMC and Samsung for the loyalty of fabless chip designers. Part of that pitch is offering a complete manufacturing service that includes not just wafer fabrication (at Intel's US and European fabs) but advanced packaging. A world-class packaging facility in Malaysia gives Intel's foundry sales team a credible end-to-end offering.

Government incentives. In 2022, Intel pledged to invest RM30 billion over a decade in Malaysia, securing significant incentive packages from MIDA. The Malaysian government's willingness to co-invest through the NSS fiscal envelope makes Malaysia financially competitive with the US CHIPS Act subsidy regime for packaging-specific work, where Intel does not qualify for the same level of US taxpayer support.

The risks attached to the Pelican investment are significant and not hypothetical. Intel's financial position has been under severe stress since 2023, with the company undergoing a major restructuring that includes workforce reductions, asset sales, and a painful transition of its foundry business toward external customer revenue. If Intel's IFS strategy fails to attract sufficient fabless customer volume, the Pelican facility's utilisation rate becomes uncertain — a \$7 billion asset running at 60% capacity is a financial liability, not an industrial asset. The other risk is competitive: TSMC has been advancing its own CoWoS and SoIC technologies aggressively, and NVIDIA, the largest potential packaging customer, has been committed to TSMC's packaging ecosystem. If Intel cannot secure major non-Intel packaging customers for Pelican, the facility serves primarily as Intel's own supply chain infrastructure — still strategically valuable, but less transformative for Malaysia's broader ambition to become a neutral advanced packaging hub.

Nevertheless, the Pelican facility represents a fundamental upgrade of Malaysia's semiconductor identity. Regardless of Intel's internal corporate trajectory, a functioning world-class advanced packaging complex on Malaysian soil — with trained engineers, supply chain relationships, and process certifications — permanently elevates Malaysia's capability ceiling in the global semiconductor industry.

Sources:

- Intel Completes "Project Pelican" Malaysia Packaging Fab — TechPowerUp
 - Intel Ramps Up Advanced Packaging: Malaysia Complex Operational in 2026 — TrendForce
 - Intel to Inject \$208M in Malaysia — TrendForce
 - HBM Shipments to Malaysia Surge, Pointing to Intel's "Project Pelican" Facility — TechPowerUp
 - TSMC Taps Intel's Malaysia Plant for CoWoS Backend Capacity Support — BigGo Finance
-

V. The Front-End Question — Can Malaysia Actually Do It?

The honest answer, delivered at SEMICON Southeast Asia 2026, was: not yet, not at scale, and perhaps not within the timeframe the government has specified. That candour from industry insiders does not invalidate Malaysia's front-end ambitions — it contextualises them.

The gap between Malaysia's OSAT mastery and credible front-end wafer fabrication capability is enormous and multidimensional.

Capital intensity. A competitive logic node front-end fab — say, 28nm CMOS — requires \$3 to 5 billion in capital investment for a greenfield facility at useful scale. A leading-edge EUV-dependent fab (7nm and below) requires \$20 billion or more. Malaysia's entire NSS fiscal envelope of RM25 billion (approximately \$5.3 billion) is roughly equivalent to the cost of one competitive but not cutting-edge 28nm fab. The government cannot afford to build a competitive front-end fab using public money alone; it must attract private capital to do so.

Process chemistry and technology know-how. Running a back-end assembly line and running a front-end semiconductor process require entirely different knowledge domains. Front-end fabrication involves hundreds of sequential process steps — thermal oxidation, ion implantation, chemical vapour deposition, physical vapour deposition, photolithography, etch, chemical mechanical polishing — each requiring tight process control, specialised engineers, and years of yield-learning. Malaysia's two existing fabs — Silterra (operating 110-180nm nodes) and Infineon Kulim (power semiconductors) — demonstrate that front-end capability exists in Malaysia, but at mature nodes and within a narrowly defined technology base. The leap to logic nodes competitive with the global foundry industry (TSMC, Samsung, GlobalFoundries, UMC) requires not incremental investment but a paradigm shift in institutional capability.

EUV access. Extreme Ultraviolet lithography, manufactured exclusively by ASML in the Netherlands, is the enabling technology for sub-7nm logic nodes and is subject to tight export controls that already restrict access for certain jurisdictions. Malaysia, as an unaligned nation with complex China ties, faces uncertainty over whether it would receive reliable EUV system supply if it attempted to build a leading-edge fab. Even at 28nm — which uses DUV multi-patterning, not EUV — the equipment investment and know-how transfer challenge is substantial.

Water and power. Front-end fabs are extraordinarily resource-intensive. A typical leading-edge fab consumes 3 to 5 million gallons of ultrapure water per day and 50 to 100 megawatts of electricity continuously. Malaysia's water availability in Penang and Kedah is adequate for current operations but has historically created supply tensions during dry seasons. Power infrastructure would require significant upgrades to support large-scale front-end expansion, though Malaysia's relatively low electricity costs and the government's commitment to renewable energy expansion under the National Energy Transition Roadmap are positive factors.

The Taiwan reference point. TSMC took approximately 40 years — from its 1987 founding to the N3 node in 2022 — to reach the manufacturing frontier. Its success required sustained government partnership (including the Industrial Technology Research Institute as TSMC's precursor), an extraordinary concentration of human capital, and a political consensus in Taiwan that semiconductors were a national strategic priority. Malaysia is not starting from zero, but it is also not starting from where Taiwan started in 1987. The institutional, workforce, and supply chain depth required for competitive front-end manufacturing has no rapid-assembly shortcut.

Given these constraints, industry analysts at SEMICON SEA 2026 identified compound semiconductors — specifically silicon carbide (SiC) and gallium nitride (GaN) — as Malaysia's most credible near-term entry point into genuine front-end manufacturing. The rationale is compelling:

Compound semiconductor fabs operate at 6-inch and 8-inch wafer diameters, compared to the 12-inch standard for leading-edge logic fabs, reducing capital requirements substantially. They do not require EUV lithography — current SiC and GaN processes use conventional photolithography tools that are commercially available without export control complications. Their device performance in power electronics (electric vehicle inverters, EV charging stations, industrial motor controllers, data centre power supplies) and radio-frequency applications (5G base stations, defence radar, satellite communications) is not replicable by conventional silicon at equivalent form factors. And their market demand is growing at rates that exceed conventional silicon — the global SiC power device market is expanding rapidly as electric vehicle adoption accelerates globally.

This is precisely the window that CHIPX is attempting to exploit. The Irish-Taiwanese compound semiconductor startup announced in December 2025 its intention to establish an 8-inch GaN-on-SiC wafer fabrication facility in Malaysia — the first facility of this type in the ASEAN region. The facility aligns with Malaysia's NSS and NIMP 2030 frameworks and targets

applications in AI data centre power management, aerospace, and high-performance computing — markets where GaN's high-voltage, high-frequency capabilities are directly relevant.

The compound semiconductor path is not a consolation prize. It represents a genuine front-end capability that Malaysia can realistically build within a five-year horizon, creating the engineering culture, equipment supplier relationships, and process knowledge that form the foundation for more ambitious moves in subsequent phases.

Sources:

- SEMICON SEA 2026: Can Malaysia's Semiconductor Industry Break Into Front End? — TechWire Asia
 - CHIPX to establish 8-inch GaN-on-SiC wafer fab in Malaysia — Semiconductor Today
 - CHIPX Plans for ASEAN region's First 200-mm Wafer Fab — Power Electronics News
 - 'No waiting on semicon wafer fabs' — The Star
 - DNeX and Foxconn — Foxconn to build chip fab in Malaysia — DCD
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VI. The JS-SEZ Semiconductor Angle: Johor as the Next Tier

The Johor-Singapore Special Economic Zone, established by bilateral agreement in January 2025 and covering 357,128 hectares across Iskandar Malaysia, Forest City, Pengerang Integrated Petroleum Complex, and Desaru, introduces a qualitatively different architecture for Malaysia's semiconductor ambitions in the southern peninsula.

The fundamental logic of the JS-SEZ is geographic arbitrage formalised as industrial policy. Singapore — operating with one of the most sophisticated IP protection regimes, capital markets, and technical talent pools in Asia — cannot cost-effectively host large-scale manufacturing. Land costs approximately SGD2,000 to 6,000 per square metre in Singapore's western industrial zones. Labour costs for semiconductor manufacturing workers are roughly three to four times higher than equivalent roles in Johor, fifty kilometres across the Causeway. The SEZ creates a juridical and logistical framework in which multinationals can base headquarters, R&D operations, and high-value engineering functions in Singapore while locating manufacturing, warehousing, and testing in Johor — accessing Singapore-quality governance and Johor-scale economics simultaneously.

For semiconductor manufacturing specifically, the JS-SEZ creates a pathway from Malaysia's existing OSAT strength toward advanced packaging and eventually chiplet integration that mirrors the trajectory TSMC has pioneered in Taiwan. The mechanism works in stages:

Stage 1 — OSAT to advanced packaging. Malaysia's existing companies — Inari, Unisem, Malaysian Pacific Industries — operate conventional wire-bonded and flip-chip assembly processes. Advanced packaging (fan-out wafer level packaging, 2.5D interposer, 3D stacking) requires higher capital density and process control but is achievable without front-end wafer fabrication equipment. The JS-SEZ's RM185 million Malaysia Advanced Packaging Consortium

(MAPC), launched in July 2026, brings together five local companies specifically to build this advanced packaging capability. The target: capture approximately 7% of the global advanced packaging market by 2035.

Stage 2 — Chiplet integration. The global semiconductor industry is migrating from monolithic chip designs to chiplet-based architectures where multiple specialised dies — compute, memory interface, I/O — are integrated in a single package. This is TSMC's CoWoS technology, Intel's EMIB/Foveros, and Samsung's I-Cube. As chiplet designs proliferate for AI processors, server CPUs, and networking chips, the packaging facility becomes the integration site where different dies (potentially from different fabs) are combined. A world-class advanced packaging facility in Johor — connected to Singapore's IP, logistics, and finance infrastructure — could position Malaysia as a neutral chiplet integration platform.

Stage 3 — Technology transfer via A*STAR. Singapore's Agency for Science, Technology and Research (A*STAR) operates semiconductor research programmes including the Institute of Microelectronics (IME), which has been involved in research on advanced packaging, photonics integration, and wide bandgap semiconductors. The JS-SEZ framework creates pathways for technology developed at A*STAR facilities in Singapore to be commercialised through manufacturing operations in Johor — a formalised knowledge transfer channel that Malaysia's northern clusters (Penang, Kulim) do not have equivalent access to.

The "queen bee" metaphor used by Malaysian officials to describe the JS-SEZ's semiconductor catalyst strategy refers to the anticipation that a marquee anchor investor — a TSMC, an ASML, or a tier-one OSAT company establishing its most advanced facility — will attract a supplier ecosystem of smaller companies that cluster around it. Johor's approved investments reached approximately US\$23 billion by Q3 2025, though not all of this is semiconductor-specific; the figure spans advanced manufacturing, digital infrastructure, and logistics. The May 2026 Micron-OCBC JS-SEZ Supplier Event at SEMICON SEA was a concrete indicator that Micron Technology — which operates major DRAM manufacturing in Singapore — is beginning to develop a Johor supplier ecosystem that could include Malaysian semiconductor services companies.

The challenge for the JS-SEZ semiconductor thesis is attracting the "queen bee" in the first place. Without a tier-one anchor — a company of TSMC's, ASML's, or Applied Materials' calibre willing to commit genuinely new advanced manufacturing capacity — the JS-SEZ risks becoming a business-friendly industrial park rather than a semiconductor cluster catalyst.

Sources:

- MIDA Anchors Malaysia As A Strategic Semiconductor Supply Chain Partner Under Johor-Singapore SEZ
- JS-SEZ: Presence of semiconductor 'queen bee' expected to create new ecosystem — NST
- RM185mil to drive advanced chip packaging — The Star
- Govt bets on five local semiconductor companies capturing 7% of global advanced packaging market by 2035 — The Edge

VII. The Workforce Crisis: Malaysia's Most Binding Constraint

If there is a single variable most likely to determine whether Malaysia's semiconductor ascent succeeds or stalls, it is not capital — the NSS and foreign investment have demonstrated that capital can be mobilised. It is human capital: the shortage of engineers, technicians, and process specialists who can design, build, operate, and maintain the semiconductor facilities that Malaysia is constructing.

The scale of the shortfall is extraordinary. Industry estimates presented at SEMICON SEA 2026 indicate that Penang alone is short approximately 50,000 semiconductor engineers. Malaysia's universities and polytechnics produce roughly 5,000 engineering graduates annually who enter the semiconductor workforce — a ratio that implies a tenfold gap between supply and demand. The NSS target of 60,000 trained high-skilled engineers represents an acknowledgement of this deficit, but the target provides a destination without a credible route map.

The talent problem has several compounding dimensions:

Brain drain to Singapore. The Singapore salary premium for semiconductor engineers is substantial — Singapore-based packaging and test engineers typically earn two to three times their Malaysian counterparts in absolute terms, and four to five times more in purchasing power parity terms when regional cost-of-living differentials are applied. Penang's semiconductor companies lose an estimated 15% of their semiconductor talent to Singapore annually, primarily to Micron, GlobalFoundries, and semiconductor equipment companies based in Woodlands, Jurong, and Ang Mo Kio. The JS-SEZ, ironically, could accelerate this dynamic: by making the Johor-Singapore commute more economical and logistically viable, it may increase the pull of Singapore employment for Malaysian engineers who were previously unwilling to relocate.

Industry-academia misalignment. Malaysia's public universities — Universiti Teknologi Malaysia (UTM), Universiti Sains Malaysia (USM), Universiti Teknologi PETRONAS (UTP), and Universiti Malaysia Perlis (UniMAP) — all have semiconductor-related engineering programmes. However, industry feedback consistently highlights a gap between academic curriculum and industry requirements. Fresh graduates often lack practical exposure to modern packaging equipment, chip design EDA tools, or the specific process chemistries relevant to current industry operations. The time required to bring a fresh engineering graduate to production-floor effectiveness in a complex semiconductor operation is typically twelve to twenty-four months — a significant lag that constrains workforce ramp rates.

Wage competition from within. The expansion of Malaysia's semiconductor industry is, paradoxically, worsening its own talent problem by creating intense demand within a shallow pool. Multinationals — Intel, Infineon, Texas Instruments, AMD — are competing with each other for the same limited pool of experienced Malaysian semiconductor engineers, driving salary escalation that narrows Malaysia's cost advantage over time. AMD's new 209,000 square-foot

engineering facility in Bayan Lepas, Penang, designed for over 1,200 employees and focused on semiconductor design, will absorb significant engineering talent from the existing pool.

The IC design talent gap. Malaysia's NSS explicitly targets chip design as a value-chain upgrade priority. The country now has 32 ARM chip design licenses distributed to companies including SkyeChip Bhd, Oppstar Bhd, and GreatAsic Technology under a RM1.1 billion national partnership with Arm Holdings signed in March 2025. The Malaysia Semiconductor IC Design Park 2 in Cyberjaya provides facility infrastructure. But chip design requires the deepest specialisation in the semiconductor talent hierarchy — digital and analogue IC designers, verification engineers, physical design engineers — and these skill profiles take five to ten years to develop from a fresh engineering graduate. Malaysia's domestic design talent pool in 2026 is characterised as "nascent" by industry observers; the 32 Arm licenses represent an aspiration, not yet a capability.

MIDA's talent retention incentives — including reduced personal income tax rates for returning semiconductor professionals under the Returning Expert Programme — provide some structural support. The sector is expected to create 12,000 to 15,000 additional technical positions by end-2026, while current pipelines are estimated to meet only 60-65% of that demand, confirming a persistent gap even at current growth rates.

Sources:

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 - Malaysia's Tech Talent Shortage — The Diplomat
 - Malaysia's Arm-backed chip push lifts SkyeChip, Oppstar — Digitimes
 - AMD Expands R&D Footprint with State-of-the-art Malaysian Facility — AMD
-

VIII. Geopolitical Positioning: Walking the Tightrope Between Washington and Beijing

Malaysia's geopolitical position in the global semiconductor industry is uniquely precarious: deeply integrated into the US-designed semiconductor architecture (US equipment, US design tools, US-based customer base), yet simultaneously maintaining political and economic ties with China that Washington regards as strategically problematic.

The tension crystallised in 2025 around a specific and verifiable data point: China's semiconductor equipment imports from Malaysia surged more than 100% year-on-year to USD3.4 billion in 2025, reaching an all-time high, while China's direct imports of US chipmaking equipment fell 34% to approximately USD2 billion. Singapore simultaneously saw a 17% increase to USD5.7 billion in equipment imports from China. The conclusion drawn by US trade officials and published by the East Asia Forum (June 2026) was unambiguous: China was systematically rerouting access to semiconductor equipment through Southeast Asian

intermediaries — primarily Malaysia and Singapore — as US export controls on direct China-to-US equipment trade tightened.

The US response was swift and applied through diplomatic channels rather than formal sanctions. Washington demanded that Malaysia closely track movement of high-end NVIDIA AI chips sold into the country, amid suspicions that processors designated for Malaysian data centres were being transhipped to China. Malaysia's Trade Minister Tengku Zafrul Aziz and Digital Minister Gobind Singh Deo jointly launched a dedicated task force to monitor semiconductor and AI chip exports and establish a registry of authorised data centre end-users. Malaysia also committed to mirroring key elements of the US AI chip export control framework.

The structural paradox is captured by the ISEAS-Yusof Ishak Institute's May 2026 analysis, "Neutrality by Sector: Malaysia's Geoeconomic Alignment under Anwar." Malaysia's nominal foreign policy position is non-alignment — a legacy of the ASEAN way and Malaysia's historical positioning as a mediator between Cold War blocs. But its semiconductor sector is already structurally aligned with the US-led technology ecosystem in a way that neutrality cannot fully describe. Applied Materials, ASML, KLA, Lam Research, Tokyo Electron — the equipment companies whose tools populate Malaysia's semiconductor facilities — are all subject to US or allied-country export controls. Malaysia's access to these tools depends on maintaining a relationship with Washington that does not trigger secondary sanctions or export licence denials.

The January 2025 Biden AI Diffusion Rule placed Malaysia in Tier 2 of a global access framework, capping the country's ability to import advanced AI GPUs at 50,000 units over two years — a constraint that limits Malaysia's data centre and AI infrastructure ambitions independently of its semiconductor manufacturing goals. The Trump administration's subsequent AI chip policy modifications adjusted some of these constraints, but Malaysia remains in a category that requires US government oversight of advanced chip deployments.

Malaysia's response to this geopolitical pressure has been to perform visible compliance actions — the task force, the data centre registry, the mirroring of export controls — while preserving the commercial relationships with Chinese companies that provide an important customer base for Malaysian semiconductor services. Unisem's Chinese ownership creates ongoing strategic ambiguity. Malaysian-Chinese joint ventures in compound semiconductor materials (the SuperSiC project identified by Digitimes in July 2025) illustrate the depth of industrial entanglement.

The long-term structural risk is a forced alignment choice. If the US-China semiconductor decoupling accelerates to the point where Washington demands that its semiconductor equipment suppliers and IP licensors remove Malaysia from their approved customer lists — as occurred with Huawei's suppliers — Malaysia's entire semiconductor industry would face an existential supply-chain crisis. This is not the current trajectory; Malaysia has demonstrated willingness to implement US-compatible export controls. But the risk exists, and Malaysia's government is clearly aware of it.

Sources:

- China's chipmaking supply chain runs through Southeast Asia — East Asia Forum
 - China snaps up US chip tools via Southeast Asia — Nikkei Asia
 - Malaysia to Tighten Chip Controls — The Diplomat
 - Neutrality by Sector — ISEAS
 - How Malaysia's Semiconductor Industry Can Thrive — The Diplomat
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IX. Superforecast: Three Scenarios to 2030

Any credible intelligence assessment must translate the strategic analysis above into probabilistic scenarios. The three scenarios below are assessed against the current evidence base as of August 2026 and are calibrated to the 2030 horizon — the end of the NSS Phase 2 timeline.

Scenario A: Front-End Breakthrough at Mature Nodes (Probability: 25%)

In this scenario, Malaysia achieves credible front-end semiconductor manufacturing capability by 2030, anchored by compound semiconductor production (8-inch GaN/SiC) at commercial scale and at least one 28nm CMOS wafer fab either operational or in advanced construction — most plausibly the DNeX-Foxconn 12-inch facility, which has been in planning since 2022 and targets 28-40nm process nodes. The compound semiconductor pathway via CHIPX's GaN-on-SiC facility provides the front-end proof of concept; the DNeX-Foxconn fab provides the logic node entry point.

For this scenario to materialise, the following conditions must hold: CHIPX successfully ramps its 8-inch GaN-SiC facility to commercial volume by 2027-2028; DNeX secures committed customer volume (likely automotive and IoT microcontrollers) that justifies proceeding with the 12-inch 28nm facility; the talent shortage is partially ameliorated by aggressive upskilling programmes and strategic immigration of engineering talent; and the geopolitical environment remains stable enough that US equipment suppliers continue to sell to Malaysian fabs without restriction.

In this scenario, Malaysia's semiconductor revenue exceeds USD25 billion by 2030, and the country's global semiconductor identity begins to shift from "OSAT colony" to "integrated compound semiconductor producer and advanced packager."

Scenario B: AI Packaging Boom Capture (Probability: 55% — Base Case)

This is the most probable outcome given current trajectories. In this scenario, Malaysia does not achieve meaningful front-end logic capability by 2030 but successfully captures a disproportionate share of the advanced packaging boom driven by AI chip demand. The advanced packaging market — encompassing 2.5D, 3D stacking, fan-out wafer-level packaging,

and chiplet integration — is growing at double-digit rates driven by AI processor, high-performance computing, and networking chip requirements. Advanced packaging offers margins of 40-50%, far above conventional OSAT margins of 10-20%.

Intel's Pelican facility becomes operational and attracts third-party customers (potentially including TSMC CoWoS overflow volume). Malaysia's five-company MAPC consortium captures the targeted 7% of the global advanced packaging market by 2035. The JS-SEZ successfully attracts two or three marquee advanced packaging anchor investors by 2028. Malaysia's semiconductor industry revenue grows to USD18-22 billion by 2030, with the advanced packaging segment representing an increasing share of total value.

In this scenario, Malaysia has not solved its front-end gap, but it has repositioned itself from "commodity OSAT" to "advanced packaging hub" — a qualitative value-chain ascent that justifies the NSS investment and protects the industry from the wage-based competition from Vietnam and Thailand that threatens commodity assembly.

Scenario C: OSAT Market Share Erosion (Probability: 20%)

In this downside scenario, Malaysia's semiconductor ambitions outpace execution capacity, and the combination of the talent shortage, geopolitical friction, and aggressive regional competition from Vietnam and Thailand erodes Malaysia's OSAT market share before the advanced packaging and design upgrades can be operationalised. Vietnam — which has set a target of 50,000 semiconductor engineers by 2030 — is moving rapidly up the learning curve, supported by significant US investment (Intel has operations in Ho Chi Minh City; Samsung is a major electronics manufacturer) and a younger demographic that provides a larger graduate pipeline. Thailand's automotive power semiconductor strategy targets SiC and GaN applications that overlap with Malaysia's compound semiconductor ambitions.

In this scenario, the 13% global OSAT market share that Malaysia has held for a decade begins to decline toward 10-11% by 2030, as new capacity additions in Vietnam, Thailand, and potentially India absorb investment that would previously have defaulted to Malaysia. The NSS front-end targets are delayed by three to five years beyond the original timeline. Advanced packaging develops but at smaller scale than targeted. Malaysia's semiconductor sector revenue grows modestly to USD13-15 billion by 2030, below the trajectory implied by NSS targets.

The risk factors that push toward this scenario include: failure of the DNeX-Foxconn fab to reach construction initiation by 2027; CHIPX GaN-SiC facility delays due to equipment access or financing; continued AI Diffusion Rule constraints limiting advanced chip access; and inability to resolve the engineering talent shortage through university reform and immigration policy.

Sources:

- Malaysia's local semiconductor players are making their advanced packaging move — CRN Asia

- Malaysia has zero advanced packaging semiconductor capability. Five companies are trying to change that — TechWire Asia
 - Why Vietnam's semiconductor strategy could outpace Malaysia's — Aliran
 - Malaysia's Semiconductor Base Can Power Its Next AI Growth Phase — BusinessToday
 - How Vietnam, Malaysia, and Singapore Are Shaping ASEAN's Semiconductor Supply Chain — ASEAN-BAC
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X. Data Sources and Methodology

All quantitative data in this report was sourced from live web queries conducted on 28 August 2026. No figures were drawn from the author's training data. BLMIM API (localhost:8742) was offline at the time of report production; all data derives from primary web sources.

Primary Sources Cited

Source	Data Provided
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MIDA	RM426.7B record investment 2025
MIDA — NSS Announcement	RM25B fiscal allocation, NSS targets
TechPowerUp	Project Pelican completion status
TrendForce, March 2026	Intel Malaysia packaging complex 2026 operations
East Asia Forum, June 2026	\$3.4B equipment rerouting data
Semiconductor Today, Dec 2025	CHIPX GaN-SiC 8-inch fab announcement
TechWire Asia, May 2026	SEMICON SEA 2026 front-end assessment
ISEAS, May 2026	Geopolitical alignment analysis
The Edge Malaysia	MAPC 7% advanced packaging market target 2035
The Star, July 2026	RM185M advanced packaging investment
Bernama	RM92.8B Q1 2026 approved investments
Intel History	Intel Penang 1972 founding history
CRN Asia	32 ARM licenses deployed
Digitimes, May 2026	SkyeChip ARM partnership

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